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Original Article Title: *Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study*

To: The Editor, *Knowledge-Based Systems*

Re: Response to Reviewers, Revision Round 1

October 5, 2026

Dear Editor and Reviewers,

We thank you for your careful and constructive evaluation of our manuscript. Your comments prompted substantive revisions to the experimental account, statistical reporting, notation, and scope of our conclusions. We respond to each comment below and indicate the corresponding changes in the manuscript.

The most consequential correction concerns the original synthetic-exposure comparison. During revision, we identified that its baseline and intervention conditions had been evaluated through different pipeline paths, including differences in the construction of the baseline-correct question set (K_0). Consequently, we no longer rely on that comparison as evidence for the previously reported intervention effect. The revised manuscript instead presents paired dose-response experiments conducted under consistent pipelines. On Qwen, reducing synthetic exposure attenuated the Generation 5–10 retention decline at the 50% reduction level. On Gemma 3, the corresponding intervention improved Generation 10 retention but did not attenuate the decline over that interval. We have revised the interpretation to preserve this distinction rather than proposing a common mechanism that the experiments do not establish.

We have also clarified the status of effective training pressure (ETP). It is now presented as an operational framework for organizing experimental conditions and their observed diagnostics, not as a validated universal scalar predictor or an architecture-independent threshold. Additional revisions distinguish controlled comparisons from descriptive contrasts, report the available seed-level uncertainty, define the notation used in the equations and figures, and state the limits of the factual-retention evaluation.

The responses that follow identify the action taken for each concern and direct the reader to the relevant part of the revised manuscript. Revised manuscript passages quoted in this letter appear in [blue](#).

Respectfully submitted on behalf of all authors,

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Editor

Revise (including language editing). Provide point-by-point response to each reviewer comment in the cover letter.

Author response:

Thank you for this guidance. We have edited the manuscript throughout for language and clarity, with particular attention to the description of experimental conditions, the distinction between

results and interpretations, and the consistency of terminology across sections. We have also moderated claims that extended beyond the tested architectures or evaluation setting.

Author action:

This letter provides a separate response to every concern raised by Reviewers 1 and 2. Each response describes the revision or, where a requested conclusion cannot be established from the available experiments, states that limitation explicitly. Locations in the revised manuscript are provided with the individual responses below.

Reviewer 1

Reviewer #1, Concern #1 (Prospective calibration of ETP):

How can effective training pressure be quantitatively calibrated and prospectively tested on an unseen backbone or dataset?

Author response:

The present experiments do not establish a transferable scalar ETP predictor or a prospective threshold for an unseen backbone or dataset. The revised manuscript distinguishes controllable design variables (nominal adapter rank, learning rate, planned training steps, and synthetic-data dose) from post-training diagnostics, including effective rank and weight drift. Under the tested protocol, we identify empirical intervals between adjacent configurations exhibiting different retention trajectories; these intervals should not be interpreted as universal numerical thresholds.

A prospective test would require specifying a calibration rule and outcome criterion before examining a held-out backbone or dataset, selecting configurations on both sides of the predicted regime boundary, and evaluating them without refitting that rule to the held-out outcomes. We describe effective training pressure as an *operational experimental framework*, not a validated universal predictor. The pre-registered G5 experiment (three single-seed cells on Qwen) provides limited exploratory evidence: retention was ordered in the expected direction for 2 of 3 pre-specified configurations. No Π threshold was reported for G5 because the threshold was not pre-specified, and the experiment does not test an unseen backbone or dataset. Prospective calibration remains future work.

Author action:

We added a concrete validation protocol for future prospective calibration in Section 6 and state explicitly that held-out prospective validation has not been performed in the present study. Section 3.5 was revised to introduce ETP as an operational framework. Section 4.4 reports the G5 exploratory results with appropriate hedging.

Location in revised manuscript: Section 3.5 (ETP as framework, revised); Section 6 (prospective calibration protocol added); Section 4.4 (G5 results).

Reviewer #1, Concern #2 (Separating confounds from pressure effects):

How are quantization, optimizer state, data order, and model-specific dynamics separated from the claimed pressure effects?

Author response:

Each potential confound is addressed through a dedicated control, now documented in Section 3.9 (“Sources of variation and experimental controls”) with Table 2 (tab:controls).

For quantization: all QLoRA experiments use 4-bit NormalFloat (NF4) with adapters in bf16. Ablation G3 reruns the same protocol with standard bf16 precision and no base-weight quantization, three seeds at each of $r = 16$ and $r = 256$. At $r = 16$, bf16 achieves Gen10 = 97.4% (76/78 per seed), while the original NF4 single-seed result was 96.2% (75/78); these use different seeds and pipelines and are not a paired comparison. At $r = 256$, bf16 achieves Gen10 = 79.1%, within 0.0 pp of the NF4 G1+G4 mean under the same seeds and matched protocol. Quantization is therefore not the primary driver of the regime separation.

For optimizer state: a fresh optimizer is instantiated at every generation; no momentum or adaptive-rate state is carried forward. This is held constant across all conditions within a backbone.

For data order: ablation G4 runs Qwen at $r = 256$ with three independent shuffle seeds (42/101/202), all other conditions identical to G1 (seeds 15/137/256). A permutation test comparing Gen10 between the two groups gives $p = 0.70$ (two-sided, $N = 3 + 3$). Because seed and data order varied simultaneously in G4, this result does not isolate order alone; it establishes that the combined variance is indistinguishable from seed-to-seed variance in G1. The combined $N = 6$ set yields Gen10 = 79.7% $\pm$ 1.7 pp.

Following the reviewer’s implicit distinction, we now separate ETP variables into two classes. *ETP-design* variables are known before training: nominal rank, learning rate, training steps, and synthetic exposure fraction. *ETP-observed* variables are measured after training: effective rank, mean absolute weight drift $d(t)$, response length, and SDI-3. The latter serve as post-hoc diagnostic proxies, not pre-specified predictors.

For the central Qwen result, the $r = 16$ condition uses three G3/bf16 runs (Gen10 = 76/78 per seed) and $r = 256$ uses six G1+G4 runs (mean 79.7% $\pm$ 1.7 pp). The 17.7 pp arithmetic gap is a descriptive observation, not a confirmatory permutation test, because the two groups differ simultaneously in rank, quantization, and data-order protocol. The cleanest rank-only comparison is the G3 paired design: same seeds, same bf16, rank the only variable; per-seed Gen10 differences are +15, +13, +15 items out of 78 (mean +18.4 pp); sign-flip $p = 0.25$ ($N = 3$).

For Gemma 3, the $r = 4$ vs. $r = 16$ contrast ($N = 5 + 5$, $|K_0| = 46$) gives $\Delta = +37.8$ pp; bilateral permutation over $\binom{10}{5} = 252$ allocations gives $p = 0.0079$ (independent seeds) or $p = 0.0625$ (paired sign-flip).

Author action:

We added Section 3.9 and Table 2 documenting all controls. The ETP-design vs. ETP-observed distinction was introduced explicitly in Section 3.5 and Table 1. The Qwen comparison now reports the descriptive 17.7 pp gap and the G3 paired result ($p = 0.25$, $N = 3$) separately. Gemma 3 reports both permutation tests with the seed-assignment caveat.

Location in revised manuscript: Section 3.5 (ETP definition); Table 1 (tab:notation); Section 3.9 (tab:controls); Section 4.2 (Qwen, G3 paired contrast); Section 4.3.1 (Gemma 3, $p = 0.0079$ or $p = 0.0625$); Section 4.6 (G3 and G4 controls).

Reviewer #1, Concern #3 (External validity of the evaluation):

Do the conclusions hold for longer answers, multi-hop questions, other domains, and larger factual evaluation sets?

Author response:

Three distinct concerns are raised: the evaluation metric, the dataset scope, and the extension to other domains.

The primary metric uses a bidirectional substring criterion, intentionally permissive to handle short-answer variation and aliasing. Configurations that generate longer responses may receive inflated retention scores. TriviaQA was chosen to isolate factual retention from reasoning or stylistic confounds; short answers minimise format-dependent evaluation noise and the substring criterion is resilient to verbosity changes. The limitation is that findings may not transfer to long-form generation, multi-hop reasoning, open-ended dialogue, or code synthesis; this scope restriction is stated explicitly in Section 6.

The held-out evaluation set contains 200 questions, of which $|K_0|$ are answered correctly at generation 0 (e.g., $|K_0| = 78$ for Qwen). This set is disjoint from the 2,000-question synthetic training stream. Scaling the evaluation set would alter the training-data composition and is deferred to future work.

Author action:

We performed a strict exact-match re-evaluation on saved predictions for all reported conditions. Under the strict scorer, absolute retention values are lower, but the direction of every regime classification is unchanged. This re-evaluation is reported in supplementary Table S2 and discussed in Section 6. The scope restriction to factoid single-answer evaluation is stated explicitly in Section 6.

Location in revised manuscript: Section 6, ll. 2094–2100 and ll. 2181–2193; supplementary Table S2; Section 3.1 (evaluation protocol).

Reviewer #1, Concern #4 (Exposure reduction: correction, pre-registered result, and mechanism):

Why does the five-percent exposure reduction stabilize Qwen but not Gemma 3, and does this generalize across ranks and seeds?

Author response:

This question led directly to identifying a pipeline inconsistency in the original submission and to a substantially revised, pre-registered experiment.

The original submission reported that a $\sim 5\%$ reduction of synthetic training examples raised retention by +9.0 and +9.4 pp for conditions C3 and C5 respectively. We subsequently identified that C1 (the baseline) and C5 (the intervention) had been produced by different evaluation pipelines, differing in the K_0 construction procedure and the data-loading code path. When C1 and C5 are re-run under an identical pipeline with paired seeds, the effect of $\sim 5\%$ example removal is only +1.9 pp, not distinguishable from zero at $n = 3$. This error was uncovered as a direct result of the reviewer’s question.

The pre-registered dose–response experiment (Bloco 7, protocol committed as [a916ebf](#) before any run) varies synthetic exposure at 0%, 10%, 25%, and 50% removal across five paired seeds on Qwen 2.5 1.5B at $r = 256$. At Gen 5, gains are uniform across doses: +3.6 pp (10%, $p = 0.009$), +3.8 pp (25%, $p = 0.003$), +4.4 pp (50%, $p < 0.001$). At Gen 10, the benefit is dose-graded: +1.8 pp at 10% ($p = 0.160$, not significant), +4.9 pp at 25% ($p = 0.005$, 95 % CI [2.5, 7.2]), and +9.2 pp at 50% ($p < 0.001$, 95 % CI [6.9, 11.6]). At 50% removal the Gen 5→10 slope

is -0.15 pp/generation versus -1.13 pp/generation at 0%, strongly attenuating progressive loss and shifting the system toward the Bounded regime.

The pre-registered prediction that 50% removal would match halving the learning rate from $\eta = 10^{-5}$ to 5×10^{-6} yielded a similar point estimate (Gen 5 retention 89.7% vs. 90.2%; Gen 10 difference $+0.4$ pp, 90 % CI $[-0.8, 1.6]$). Formal equivalence was not established.

A parallel pre-registered dose-response on Gemma 3 1B IT at $r = 10$ (G2, three paired seeds, $|K_{0,G2}| = 44$) shows that 50% removal raises Gen 10 retention by $+12.9$ pp ($p = 0.001$, 95 % CI $[9.70, 16.10]$). However, the Gen 5→10 slope in the G2 50% arm (-0.91 pp/generation) equals that of the 0% arm (-0.91 pp/generation) within rounding: the exposure effect on Gemma 3 manifested as a Gen 10 retention gain without reducing the rate of progressive loss. The mechanism was not established; a steps-matched control was inconclusive at $n = 3$.

Author action:

We acknowledged the pipeline inconsistency explicitly in the revised manuscript and archived the original C2–C5 results in the replication repository for transparency. Section 4.5 was rewritten to present the pre-registered B7 dose-response results for both Qwen (five seeds per dose) and Gemma 3 G2 (three seeds, 50% removal), with paired t -tests and 95 % confidence intervals. The learning-rate equivalence prediction is reported with its 90 % CI and without claiming formal equivalence.

Location in revised manuscript: Section 4.5 (B7 and G2 dose-response; pre-registration; LR equivalence); Section 6 (mechanism limitation); replication repository (original C1–C5 archived).

Reviewer #1, Concern #5 (Reproducibility package):

What complete code, checkpoints, prompts, seeds, and raw outputs will support independent reproduction?

Author response:

Independent reproduction requires more than aggregate retention tables. The reproducibility section (Section A.1) has been expanded to specify the exact model checkpoint identifiers and revisions for all three backbones; train and evaluation question IDs, and the baseline-correct K_0 sets ($|K_0| = 78$ for Qwen; $|K_{0,\text{rank}}| = 46$ and $|K_{0,G2}| = 44$ for Gemma 3, see Section 3.6); prompt and chat templates used for synthetic data generation and deterministic evaluation; distinct decoding settings (`do_sample=False` and `temperature=0` for evaluation); correctness normalization and alias-expansion code; training configurations, software versions, random seeds, data-order and downsampling masks; per-generation retention counts and boolean correctness vectors included in the consolidated CSV; and scripts to regenerate every table and figure.

Author action:

The replication package contains all analysis and figure-generation scripts; the consolidated results CSV (G1–G5, 27 executed training runs, per-generation per-seed retention); all 11 publication figures at 300 dpi; and the pre-registration commit for B7 (a916ebf). The package does *not* contain raw token-level predictions, adapter checkpoints (available on request; ≈ 70 MB each in bf16), or base-model weights. It is archived at <https://doi.org/10.5281/zenodo.23146342>. Release v1.0.3 contains the initial replication scripts and data; revised analysis scripts will be deposited in a subsequent release. Running `python scripts/make_all.py` reproduces all 112 canonical numbers (15/15 checks pass).

Location in revised manuscript: Section A.1 (repository and data access, expanded).

Reviewer 2

Reviewer #2, Concern #1 (Abbreviations and notation):

ETP appears in Figure 1 without explicit definition. MTL D is not expanded. KL, JS, and coverage appear in Figure 1 without definition. bf16, SVD, TCE, QLoRA, and LoRA lack consistently expanded forms. “Homeo.” is used in Table 2 without definition. K_0 is used both as a set and as a retention metric. T , $D_{1,T}$, D_{10} , ΔI , and related symbols in Eq. 6 are not fully defined.

Author response:

Each notation issue has been corrected. The revised manuscript does not use “ETP” as a standalone acronym; the term is introduced in full at l. 72 and used in full throughout. Figure 1 no longer contains the abbreviation in isolation. MTL D now reads “[Measure of Textual Lexical Diversity \(MTLD\)](#)” at first occurrence. KL, JS, and coverage were conceptual labels for diagnostics never computed or reported; they have been removed from Figure 1 entirely, and the revised caption describes only quantities that are measured. All acronyms now have explicit parenthetical expansions at first use: “[full fine-tuning \(FFT\)](#)” at l. 75, “[singular value decomposition \(SVD\)](#)” at l. 332, “[Quantized Low-Rank Adaptation \(QLoRA\)](#)” at l. 333. TCE (Truncated Cross-Entropy) is cited in the Related Work section as a mitigation method from prior literature; it does not appear in the Methods or Results. “Homeo.” has been expanded to “[Homeostatic](#)” in all four occurrences in Table 2.

A dedicated notation table (Table 1) distinguishes the two uses of K_0 : K_0 denotes the *set* of held-out questions answered correctly at generation 0; $|K_0|$ denotes its cardinality; and $R(t) = |K_t|/|K_0|$ denotes the retention ratio. All symbols in Eq. 6 (SDI-3) are defined in Table 1 and in the inline text preceding the equation: T = evaluation horizon in generations; $D_{1,t}$ = Distinct-1 unigram diversity at generation t ; ΔI = $\text{Instability}(T) - \text{Instability}(1)$, where $\text{Instability}(t)$ is the fraction of training items whose normalized response differs between generation $t-1$ and generation t ; MeanLen_t = mean response length in words at generation t .

Author action:

We removed “ETP” from Figure 1, expanded MTL D and all acronyms at first use, removed KL/JS/coverage from Figure 1, expanded “Homeo.” throughout Table 2, added Table 1 (tab:notation) with all symbol definitions, and added inline definitions for all Eq. 6 quantities in the preceding prose.

Location in revised manuscript: Abstract l. 72; Figure 1 caption ll. 692–698; Table 1 (tab:notation) ll. 438–445; Equations 1, 3, 5, 6; l. 875 (MTLD); ll. 75, 332, 333 (acronyms); Table 2 regime column.

Reviewer #2, Concern #2 (Equations and tables):

Equations 1–6 are severely malformed in the PDF. Eq. 1 uses “Retention.t /” instead of a clear function notation. Eq. 2 for effective rank is missing the negative sign and summation. Eq. 6 for SDI-3 has unclear signs and terms. “ 5×10^6 ” in Table 7 appears to be a rendering error.

Author response:

The malformed rendering was a font-encoding artefact: Unicode characters in mathematical mode were not mapped correctly to the T1 font metric, producing garbled output while the source was syntactically correct. The revised submission uses the `lmodern` package with T1 encoding throughout, which eliminates all rendering artefacts.

Eq. 1 now reads correctly:

$$R(t) = \frac{|\{q \in K_0 : q \text{ correct at gen. } t\}|}{|K_0|}.$$

Eq. 2 now includes the negative sign and the explicit normalisation:

$$\text{erank}(BA) = \exp\left(-\sum_i \hat{\sigma}_i \log \hat{\sigma}_i\right),$$

where $\hat{\sigma}_i = \sigma_i / \sum_j \sigma_j$. This is the standard effective-rank definition of [?].

Eq. 6 now reads:

$$\text{SDI-3} = \log \frac{\text{MeanLen}_T}{\text{MeanLen}_0} + \log \frac{\text{D1}_0}{\text{D1}_T} + \Delta I.$$

A positive SDI-3 score indicates cumulative distributional deterioration. The value in Table 7 is the learning rate 5×10^{-6} ; the exponent -6 was dropped by the same encoding issue and is now rendered correctly.

Author action:

We added `lmodern` with T1 encoding to the preamble, rewrote Equations 1, 2, and 6 with explicit notation consistent with Table 1, and verified that no other table or equation contains a similar rendering failure in the revised submission.

Location in revised manuscript: Preamble (lmodern + T1); Eq. 1 l. 595; Eq. 2 ll. 798–803; Eq. 6 ll. 881–886; Table 7 (tab:ranklr) l. 1473.

Reviewer #2, Concern #3 (Technical and data inconsistencies):

(a) Abstract and Section 4.2.3 state the Qwen boundary is between effective ranks 50–88; Section 5.1 states Qwen remains outside the degradative regime “up to effective rank 150,” contradicting Table 2. (b) $r = 128$ is classified as Bounded in Table 2 but called “above-threshold” in Section 4.4.3. (c) $r = 256$ is repeatedly called a “boundary configuration” but Table 2 classifies it as Degradative. (d) Section 5.2 states “ $r \geq 16$ on Gemma 3” but Table 4 shows $r \in \{10, 12, 14\}$ already degradative at Gen 5. (e) Abstract overclaims “three to five independent seeds per condition” when many conditions use a single seed. (f) FFT on Gemma 3 is “confirmed” from a single-seed non-monotonic result. (g) C5 removal percentage discrepancy: 15% examples vs. $\sim 5\%$ tokens.

Author response:

Each inconsistency has been corrected. The phrase “up to effective rank 150” has been removed; all threshold language now uses “an abrupt, threshold-like transition within the tested grid”, reflecting empirical localisation between adjacent tested configurations.

“Above-threshold configurations” in Section 4.4.3 now refers exclusively to $r = 256$ (Degradative). The $r = 128$ configuration (Bounded) is described as a “transition-zone configuration” where

distributional signatures begin to emerge without producing the full degradative trajectory. The five occurrences of “boundary configuration” referring to $r = 256$ have been replaced with “lowest-pressure degradative configuration tested”.

Table 4 reports retention of 78.3 %, 73.9 %, and 69.6 % for $r = 10, 12$, and 14 at Gen 5, all Degradative. The text now reads “the Gemma 3 regime boundary lies between effective ranks 3 and 6, below $r = 10$ ”; “ $r \geq 16$ ” has been replaced with “ $r \geq 10$ on Gemma 3”.

The abstract now reads “three to five independent seeds for headline conditions (single seed for intermediate ablations)”, with a dedicated N column added to all result tables. For FFT on Gemma 3, “confirms” has been replaced with “is consistent with” and the non-monotonic pattern (78.3%, 54.3%, 63.0%) is acknowledged explicitly. The 15%/5% discrepancy is clarified: C5 removed 15% of training *examples*, reducing the token count by $\sim 5\%$ because responses vary in length. The more substantive resolution is that the C1/C5 pipeline inconsistency renders the original comparison uninformative; the B7 dose–response replaces it.

Author action:

We removed “up to effective rank 150,” replaced all five occurrences of “boundary configuration” for $r = 256$, corrected the $r = 128$ label, updated the Gemma 3 boundary description, corrected the abstract seed-count claim, softened the FFT Gemma 3 conclusion, and clarified the example-vs.-token distinction for C5.

Location in revised manuscript: Abstract ll. 69, 72; Contribution #1 l. 95; Section 4.4.3 ll. 1061, 1570; ll. 1737, 1915, 2063, 2069, 2159; Section 5.2; Table 4 ll. 1265, 1277–1280; ll. 1147–1149 (FFT Gemma 3); Section 3.7 (C5 doses); Section 4.5 ll. 1708–1713.

Reviewer #2, Concern #4 (Methodology):

No formal statistical tests or confidence intervals are reported. The ETP framework is not quantified; it is a conceptual label. Cross-architecture normalisation failed with no alternative proposed. “Effective training pressure” is not a new quantitative measure. Gemma 4 E2B lacks a citation. The “sharp transition” claim is weakened by sparse rank grids.

Author response:

The revised manuscript adds seed-level exact permutation tests for all comparisons where independent training runs are available. For Qwen, the 17.7 pp arithmetic gap between $r = 16$ and $r = 256$ is reported as a descriptive observation because the two groups are not exchangeable under a rank-only null. The cleanest within-protocol comparison is G3 paired design: mean +18.4 pp, sign-flip $p = 0.25$ ($N = 3$). For Gemma 3, bilateral permutation $p = 0.0079$ (independent seeds) or sign-flip $p = 0.0625$ (paired). For dose–response, paired t -tests with 95 % CIs are reported per dose level.

ETP is presented as an *operational experimental framework*, not a validated scalar predictor. We distinguish ETP-design variables (rank, LR, exposure fraction, steps) from ETP-observed diagnostics (effective rank, $d(t)$, response length, SDI-3). The $\text{erank} \times \text{LR}$ index is labelled as exploratory throughout. The failed normalisations in Table 6 are retained as evidence of incommensurability; architecture-specific calibration is explicitly deferred to future work. The contribution is the factorial design, the dose–response evidence, the pre-registered exposure result, and the identification of architecture-specific regime boundaries.

The Gemma 4 E2B backbone is now cited as ?] (Gemma Team, 2026, arXiv:2607.02770) at every introduction point. “Sharp transition” has been replaced throughout with “[abrupt, threshold-like transition within the tested grid](#)”.

Author action:

We added permutation tests for Qwen G3 and Gemma 3, paired t -tests with CIs for all B7 and G2 dose levels, revised Section 3.5 to formalise the ETP-design/ETP-observed distinction, added the Gemma 4 citation to Sections 3.3 and 4.3, and replaced “sharp transition” throughout.

Location in revised manuscript: Section 4.2 (G3 paired contrast, $p = 0.25$); Section 4.3.1 (Gemma 3, $p = 0.0079/p = 0.0625$); Section 4.5 (dose-response, paired t -tests); Section 3.5 (ETP framework); Table 1 (tab:notation); Section 6 (cross-architecture limitation); Section 3.3 (Gemma 4 citation); Abstract l. 72.

Reviewer #2, Concern #5 (Format and citations):

References [12] and [33] share arXiv:2408.14572. Reference [32] lists year 2024 but the arXiv ID is from December 2025. No citation for Gemma 4 E2B. Keisha et al. is a preprint with no venue. Reference formatting is inconsistent.

Author response:

The arXiv number 2408.14572 now appears exactly once, assigned to ?] (*CURLoRA*, Fawi, 2024); the duplicate entry has been removed. The entry `ding2024ranktrade` (Rathore et al.) now carries `year = {2025}`, consistent with its December 2025 arXiv submission date and its appearance in AACL-IJCNLP 2025. The Gemma 4 E2B backbone is now cited as ?] at every introduction and description point. ?] is a non-peer-reviewed preprint (arXiv:2509.04796), now declared at first citation: “[? \] \(non-peer-reviewed preprint\) report...](#)”; no quantitative claim depends exclusively on it; ?] (published in *Nature*) is added as the primary citation for the model-collapse phenomenon.

Author action:

We audited and harmonised the full reference list: published works include venue, volume, pages, and DOI; arXiv-only works are formatted as `arXiv preprint arXiv:XXXX.XXXXX`. Entries `ding2024ranktrade`, `fawi2024curlora`, and `zibakhsh2024` have been completed. The full revised bibliography is provided as `cas-refs.bib` in the submission package.

Location in revised manuscript: References (cas-refs.bib): entries `fawi2024curlora`, `ding2024ranktrade`, `gemma4_2026`, `keisha2025`, all entries harmonised.